Wetland Cover Change

```
tinytex::tlmgr_install(c("fvextra", "xcolor", "framed"))
```

library

```
library(ggplot2)
library(readr)
library(dplyr)
library(janitor)
```

import data

```
Wetland.Cover.Change.LRR <- read.csv("~/Documents/Wetland Research Program/Wetland Ri
ver Rapport Indicator/Wetland Cover Change LRR.csv")
```

```


library(knitr)
library(kableExtra)

knitr::kable(Wetland.Cover.Change.LRR, format = "latex", booktabs = TRUE) %>%
  kable_styling(latex_options = "scale_down")
```

name data

```
data=Wetland.Cover.Change.LRR
```

#Local - Coastal Wetland

Plot out the change detection studies that looked at individual wetlands

filter out local data

```
local_data <- data %>%
  filter(Scale == "Local- Coastal Wetland")
```

Set arrow position just outside plot (left side)

```
arrow_x <- min(local_data$mean.year) -25
arrow_height <- 0.4
```

alternative palette

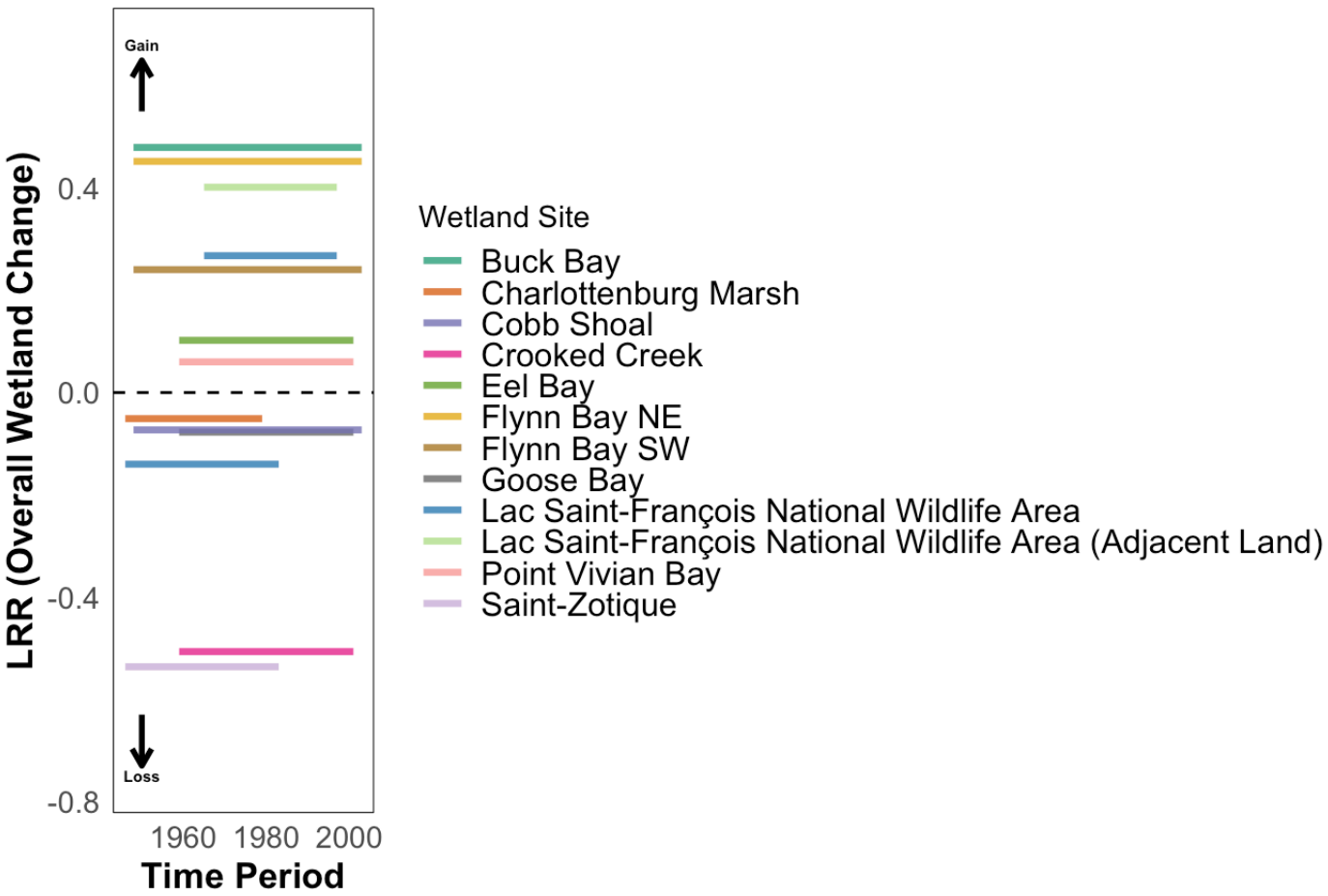
```
my_cols <- c(
  "#1B9E77", # teal
  "#D95F02", # orange
  "#7570B3", # purple
  "#E7298A", # pink
  "#66A61E", # green
  "#E6AB02", # mustard
  "#A6761D", # brown
  "#666666", # gray
  "#1F78B4", # blue
  "#B2DF8A", # light green
  "#FB9A99", # light red
  "#CAB2D6"  # lavender
)
```

plot no arrows

```
a=ggplot(local_data, aes(x = mean.year, y = LRR..ln.A..A.., color = Site)) +

  # Study duration lines
  geom_segment(
    aes(
      x = start.date,
      xend = end.date,
      y = LRR..ln.A..A..,
      yend = LRR..ln.A..A..
    ),
     linewidth = 1.2,alpha = 0.8
  ) +


  # Zero reference line
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +  scale_color_man
ual(values = my_cols)+


  # Axis + title labels
  labs(
    x = "Time Period",
    y = "LRR (Overall Wetland Change)",
    title = " ",
    color = "Wetland Site"
  ) +  annotate(
    "segment",
    x = 1950, xend = 1950,
    y = 0.55, yend = 0.65,
```

```
    arrow = arrow(length = unit(0.25, "cm")),
    linewidth = 1,
    color = "black"
  ) +

  annotate(
    "text",
    x = 1950,
    y = 0.68,
    label = "Gain",
    fontface = "bold",
    size = 2
  ) +

  # ----------------------------
  # Loss arrow (BOTTOM, outside plot)
  # ----------------------------
  annotate(
    "segment",
    x = 1950, xend = 1950,
    y = -0.63, yend = -0.73,
    arrow = arrow(length = unit(0.25, "cm")),
    linewidth = 1,
    color = "black"
  ) +

  annotate(
    "text",
    x = 1950,
    y = -0.75,
    label = "Loss",
    fontface = "bold",
    size = 2
  ) +

  # Allow drawing outside plot area
  coord_cartesian(clip = "off") +

  # Add extra space above + below
 theme_minimal() +theme(
  plot.margin = margin(10, 10, 10, 10)
)+ theme(
    panel.border = element_rect(color = "black", fill = NA, linewidth = 0.2)
  )+theme(legend.position = "right")+theme(
panel.grid = element_blank()
)+theme(strip.text = element_text(size = 14, face = "bold"))+theme(
  axis.title = element_text(size = 13, face = "bold"),
```

```
  axis.text  = element_text(size = 11))+theme(
  legend.key.height = unit(0.4, "cm"),
  legend.text = element_text(size = 12))
a
```



Regional

```
Wetland.Cover.Change.LRR_noNA <- read.csv("~/Documents/Wetland Research Program/Wetla
nd River Rapport Indicator/Wetland Cover Change LRRnoNA.csv")
```

name data

```
data=Wetland.Cover.Change.LRR_noNA
```

Plot out the change detection studies that looked at regional (river sections) coastal areas

filter out local data

```
regional_data <- data %>%
  filter(Scale == "Regional - Coastal Wetlands")
```

Set arrow position just outside plot (left side)

```
arrow_x <- min(regional_data$mean.year) -25
arrow_height <- 0.4
```

```
my_cols <- c(
  "#1B9E77", # teal
  "#D95F02", # orange
  "#E7298A", # pink
  "#66A61E", # green
  "#E6AB02", # mustard
  "#A6761D", # brown
  "#666666", # gray
  "#1F78B4", # blue
  "#B2DF8A", # light green
  "#FB9A99", # light red
  "#CAB2D6"  # lavender
)
```

reorder

```
regional_data$River.Section=factor(regional_data$River.Section,levels=c("USLR",  "Tho
usand Islands and Brockville",
  "Brockville",    "Lake St. Lawrence and Akwesasne/Cornwall","Akwesasne/Cornwall and
Lac Saint-François",
  "Lac Saint-François",
   "LSLR",
  "Lac Saint-Louis",
  "Montreal East","Montreal East/Contrecoeur",
  "Contrecoeur"))
```

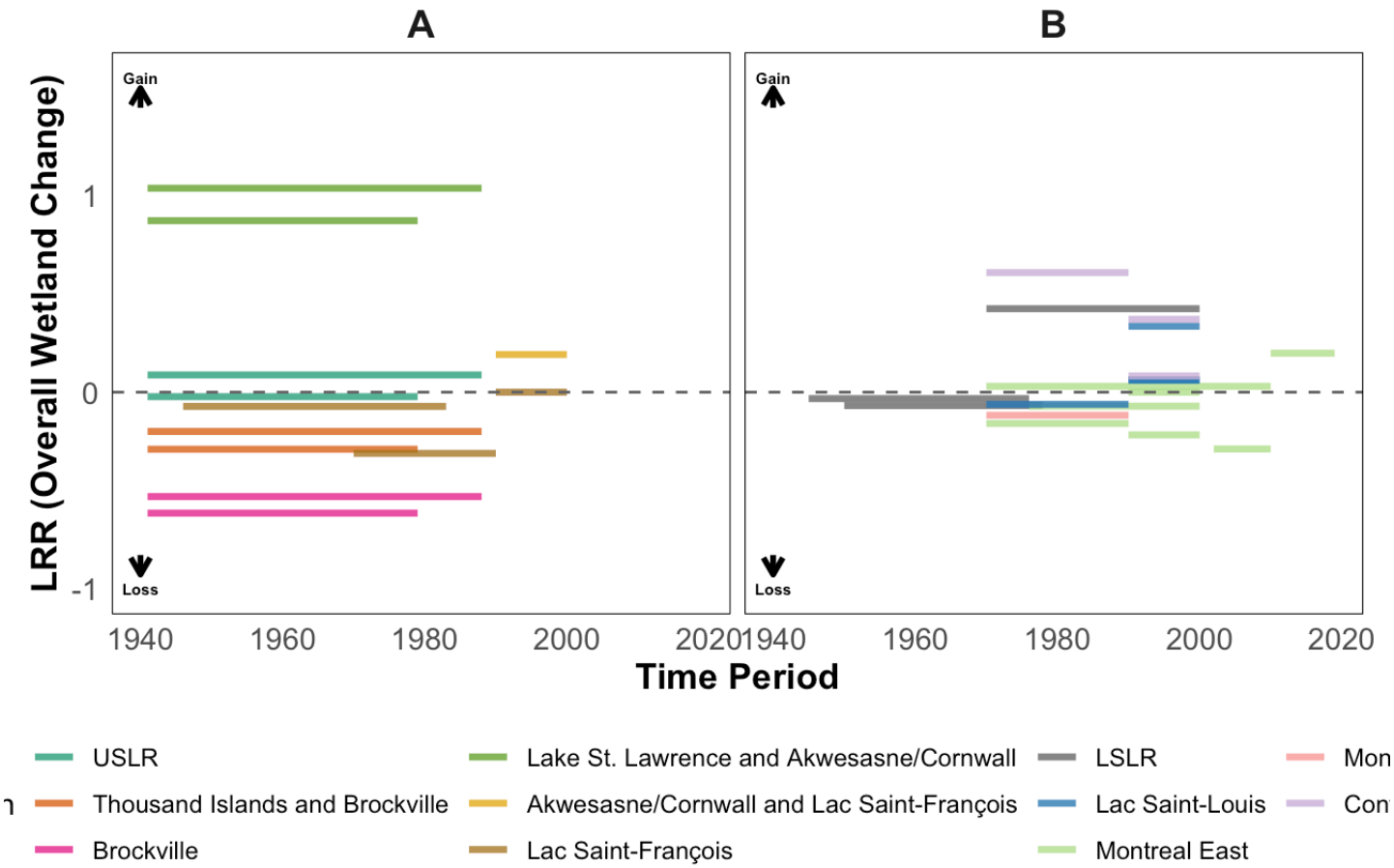
```
regional_data$Project=factor(regional_data$Project,levels=c("USLR", "KEPO"))
```

plot no arrows

```
r=ggplot(regional_data, aes(x = mean.year, y = LRR..ln.A..A.., color = River.Sectio
n)) +

  # Study duration lines
```

```
  geom_segment(
    aes(
      x = start.date,
      xend = end.date,
      y = LRR..ln.A..A..,
      yend = LRR..ln.A..A..
    ),
    linewidth = 1.2,alpha = 0.8
  ) +


  # Zero reference line
 geom_hline(yintercept = 0,
           linetype = "dashed",
           linewidth = 0.5,
           color = "gray40")+ scale_color_manual(values = my_cols)+


  # Axis + title labels
  labs(
    x = "Time Period",
    y = "LRR (Overall Wetland Change)",
    title = " ",
    color = "River Section"
  ) +  annotate(
    "segment",
    x = 1940, xend = 1940,
    y = 1.45, yend = 1.55,
    arrow = arrow(length = unit(0.25, "cm")),
    linewidth = 1,
    color = "black"
  ) +

  annotate(
    "text",
    x = 1940,
    y = 1.6,
    label = "Gain",
    fontface = "bold",
    size = 2
  ) +

  # ----------------------------
  # Loss arrow (BOTTOM, outside plot)
  # ----------------------------
  annotate(
    "segment",
```

```
    x = 1940, xend = 1940,
    y = -0.83, yend = -0.93,
    arrow = arrow(length = unit(0.25, "cm")),
    linewidth = 1,
    color = "black"
  ) +

  annotate(
    "text",
    x = 1940,
    y = -1,
    label = "Loss",
    fontface = "bold",
    size = 2
  ) + facet_wrap(~Panel)+

  # Allow drawing outside plot area
  coord_cartesian(clip = "off") +

  # Add extra space above + below
  theme_minimal() +theme(
  plot.margin = margin(10, 10, 10, 10)
)+ theme(
    panel.border = element_rect(color = "black", fill = NA, linewidth = 0.2)
  )+theme(legend.position = "bottom")+theme(
panel.grid = element_blank()
)+theme(strip.text = element_text(size = 14, face = "bold"))+theme(
  axis.title = element_text(size = 13, face = "bold"),
  axis.text  = element_text(size = 11))+theme(
  legend.key.height = unit(0.4, "cm"),
  legend.text = element_text(size = 9))
r
```



#watershed

Plot out the change detection studies that looked at regional (river sections) coastal areas

filter out local data

```
watershed_data <- data %>%
  filter(Scale == "Watershed")
```

Set arrow position just outside plot (left side)

```
arrow_x <- min(watershed_data$mean.year) -25
arrow_height <- 0.4
```

set colours

```
my_cols<- c(
  "#1B9E77", # teal
  "#D95F02", # orange
  "#7570B3", # purple
  "#E7298A", # pink
  "#66A61E", # green
  "#E6AB02", # mustard
  "#A6761D", # brown
  "#666666", # gray
  "#1F78B4", # blue
  "#B2DF8A", # light green
  "#FB9A99", # light red
  "#CAB2D6", # lavender
  "#17BECF"  # cyan (NEW)
)
```

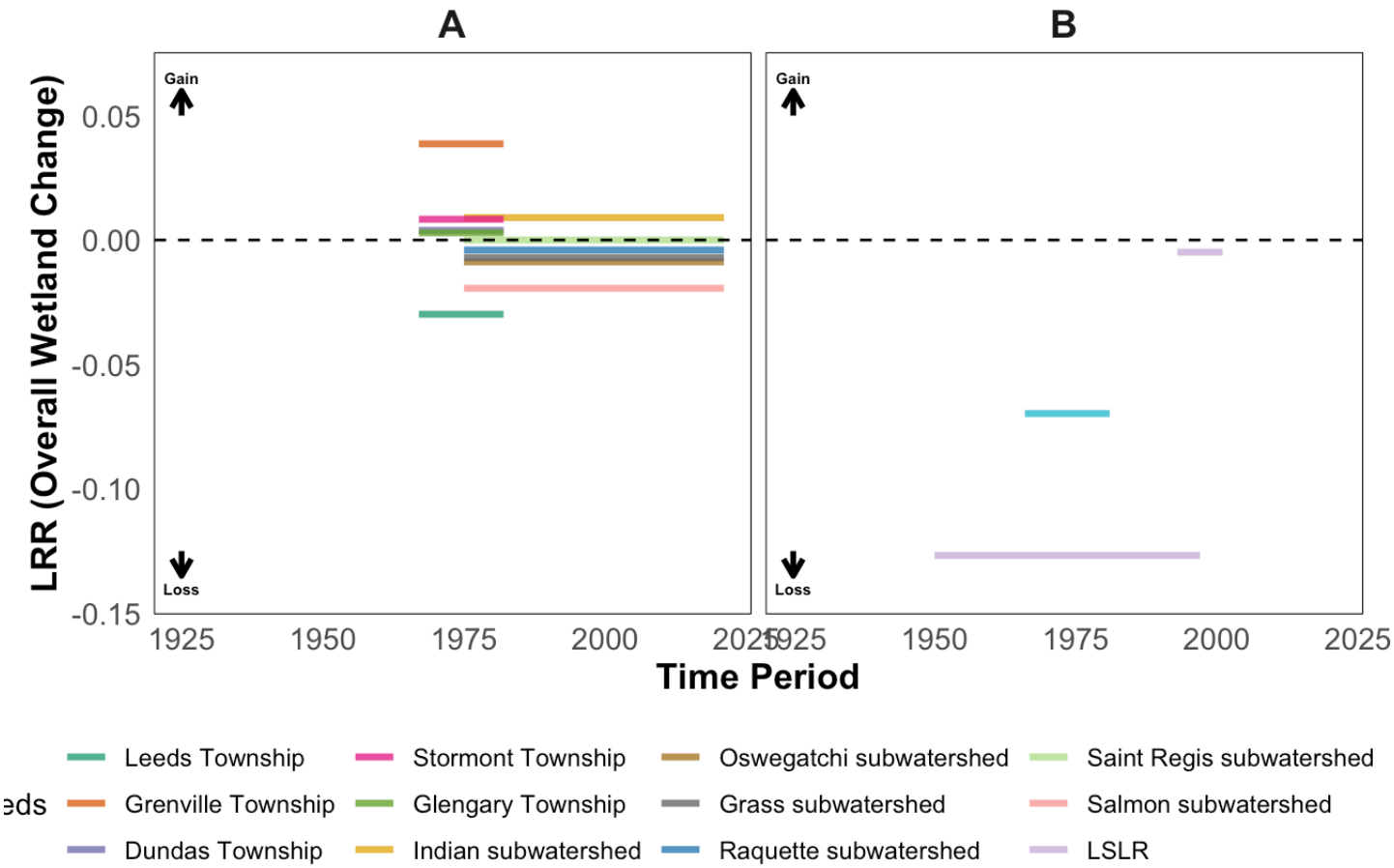
reorder

```
watershed_data$River.Section=factor(watershed_data$River.Section,levels=c("Leeds Town
ship",
"Grenville Township",
"Dundas Township",
"Stormont Township",
"Glengary Township", "Indian subwatershed","Oswegatchi subwatershed",
"Grass subwatershed",
"Raquette subwatershed",
"Saint Regis subwatershed",
"Salmon subwatershed",
"LSLR",
"Kahnawake Region of Interest"
))
```

```
w=ggplot(watershed_data, aes(x = mean.year, y = LRR..ln.A..A.., color = River.Sectio
n)) +

  # Study duration lines
  geom_segment(
    aes(
      x = start.date,
      xend = end.date,
      y = LRR..ln.A..A..,
      yend = LRR..ln.A..A..
    ),
    linewidth = 1.2,alpha = 0.8
  ) +
```

```
  # Zero reference line
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +  scale_color_man
ual(values = my_cols)+


  # Axis + title labels
  labs(
    x = "Time Period",
    y = "LRR (Overall Wetland Change)",
    title = " ",
    color = "Portions of Watersheds"
  ) +  annotate(
    "segment",
    x = 1925, xend = 1925,
    y = 0.05, yend = 0.06,
    arrow = arrow(length = unit(0.25, "cm")),
    linewidth = 1,
    color = "black"
  ) +

  annotate(
    "text",
    x = 1925,
    y = 0.065,
    label = "Gain",
    fontface = "bold",
    size = 2
  ) +

  # ----------------------------
  # Loss arrow (BOTTOM, outside plot)
  # ----------------------------
  annotate(
    "segment",
    x = 1925, xend = 1925,
    y = -0.125, yend = -0.135,
    arrow = arrow(length = unit(0.25, "cm")),
    linewidth = 1,
    color = "black"
  ) +

  annotate(
    "text",
    x = 1925,
    y = -0.14,
```

```
    label = "Loss",
    fontface = "bold",
    size = 2
  ) +

  # Allow drawing outside plot area
  coord_cartesian(clip = "off") + facet_wrap(~Panel)+

  # Add extra space above + below
  theme_minimal() +theme(
  plot.margin = margin(10, 10, 10, 10)
)+ theme(
    panel.border = element_rect(color = "black", fill = NA, linewidth = 0.2)
  )+theme(legend.position = "bottom")+theme(
panel.grid = element_blank()
)+theme(strip.text = element_text(size = 14, face = "bold"))+theme(
  axis.title = element_text(size = 13, face = "bold"),
  axis.text  = element_text(size = 11))+theme(
  legend.key.height = unit(0.4, "cm"),
  legend.text = element_text(size = 9))

w
```

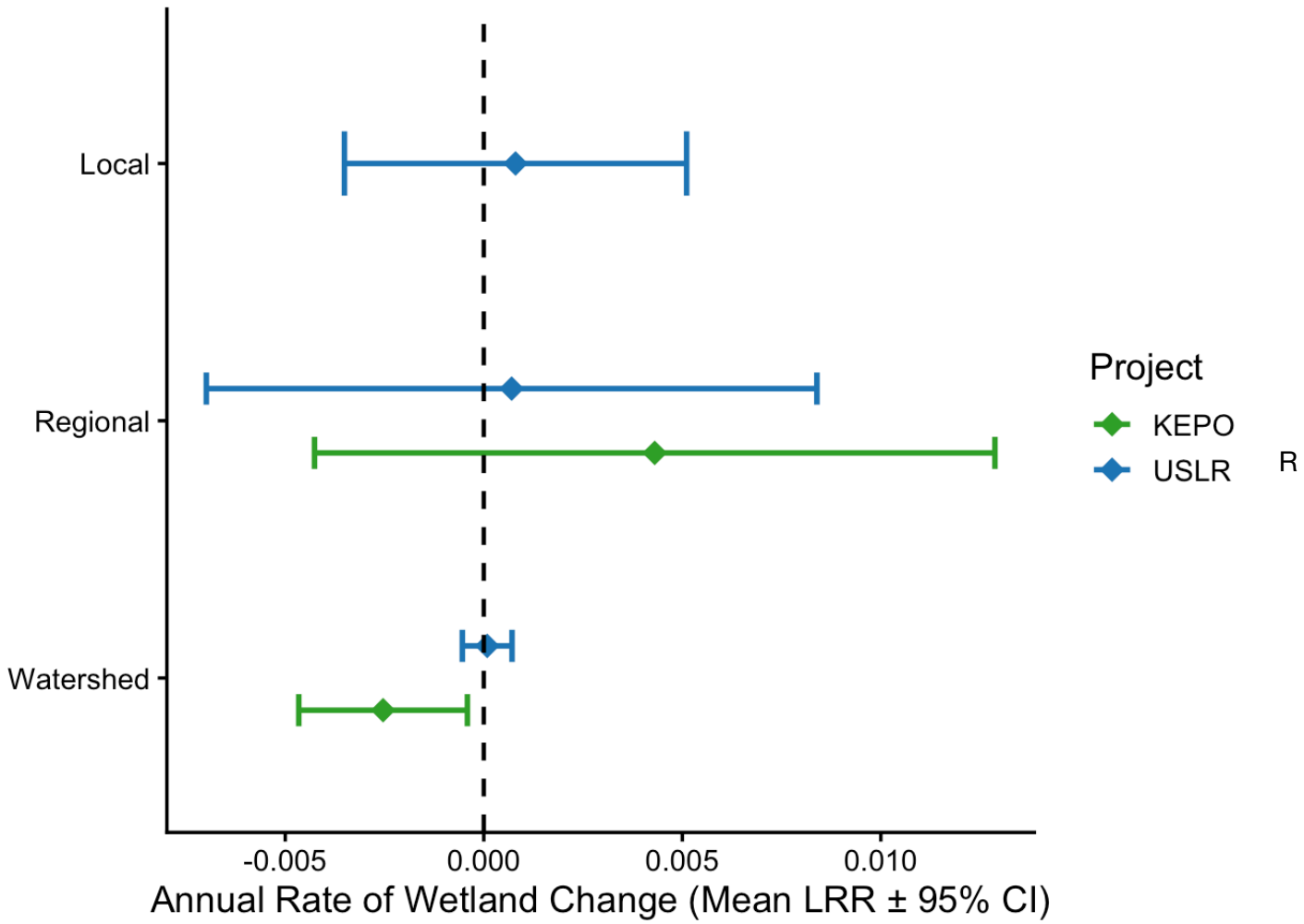


#Wetland Change Comparison of Different Scales

```
library(ggplot2)
library(dplyr)
library(readr)


# Read the CSV properly
wetland_df <- Wetland.Cover.Change.LRR_noNA



# Desired order (TOP → BOTTOM in plot)
scale_order <- c("Local- Coastal Wetland",
                 "Regional - Coastal Wetlands",
                 "Watershed")

filtered_df <- wetland_df %>%
  dplyr::filter(Scale %in% scale_order) %>%
  mutate(
```

```
    Scale = factor(Scale, levels = rev(scale_order)),
    Project = factor(Project, levels = c("USLR", "KEPO"))
  )



# Filter + reorder factor
filtered_df <- wetland_df %>%
  dplyr::filter(Scale %in% scale_order) %>%
  mutate(
    Scale = factor(Scale, levels = rev(scale_order))  # <-- reverse fixes ordering
  )

summary_df <- filtered_df %>%
  group_by(Scale, Project) %>%
  summarise(
    n = sum(!is.na(Annual.rate)),
    mean_rate = mean(Annual.rate, na.rm = TRUE),
    sd_rate   = sd(Annual.rate, na.rm = TRUE),

    se_rate   = sd_rate / sqrt(n),

    ci_lower  = mean_rate - 1.96 * se_rate,
    ci_upper  = mean_rate + 1.96 * se_rate,
    .groups = "drop"
  )

x = ggplot(summary_df, aes(x = mean_rate, y = Scale, color = Project)) +

  geom_errorbarh(
    aes(xmin = ci_lower, xmax = ci_upper),
    height = 0.25,
    linewidth = 1,
    position = position_dodge(width = 0.5)
  ) +

  geom_point(
    size = 4,
    shape = 18,
    position = position_dodge(width = 0.5)
  ) +

  geom_vline(
    xintercept = 0,
    linetype = "dashed",
    linewidth = 0.8
  ) +
```

```
  scale_color_manual(
    values = c(
   "USLR" = "#1f78b4",
"KEPO" = "#33a02c"
    )
  ) +

  scale_y_discrete(labels = c(
    "Local- Coastal Wetland"      = "Local",
    "Regional - Coastal Wetlands" = "Regional",
    "Watershed"                   = "Watershed"
  )) +

  labs(
    x = "Annual Rate of Wetland Change (Mean LRR ± 95% CI)",
    y = "",
    color = "Project"
  ) +

  theme_classic(base_size = 14)
x
```



code anova

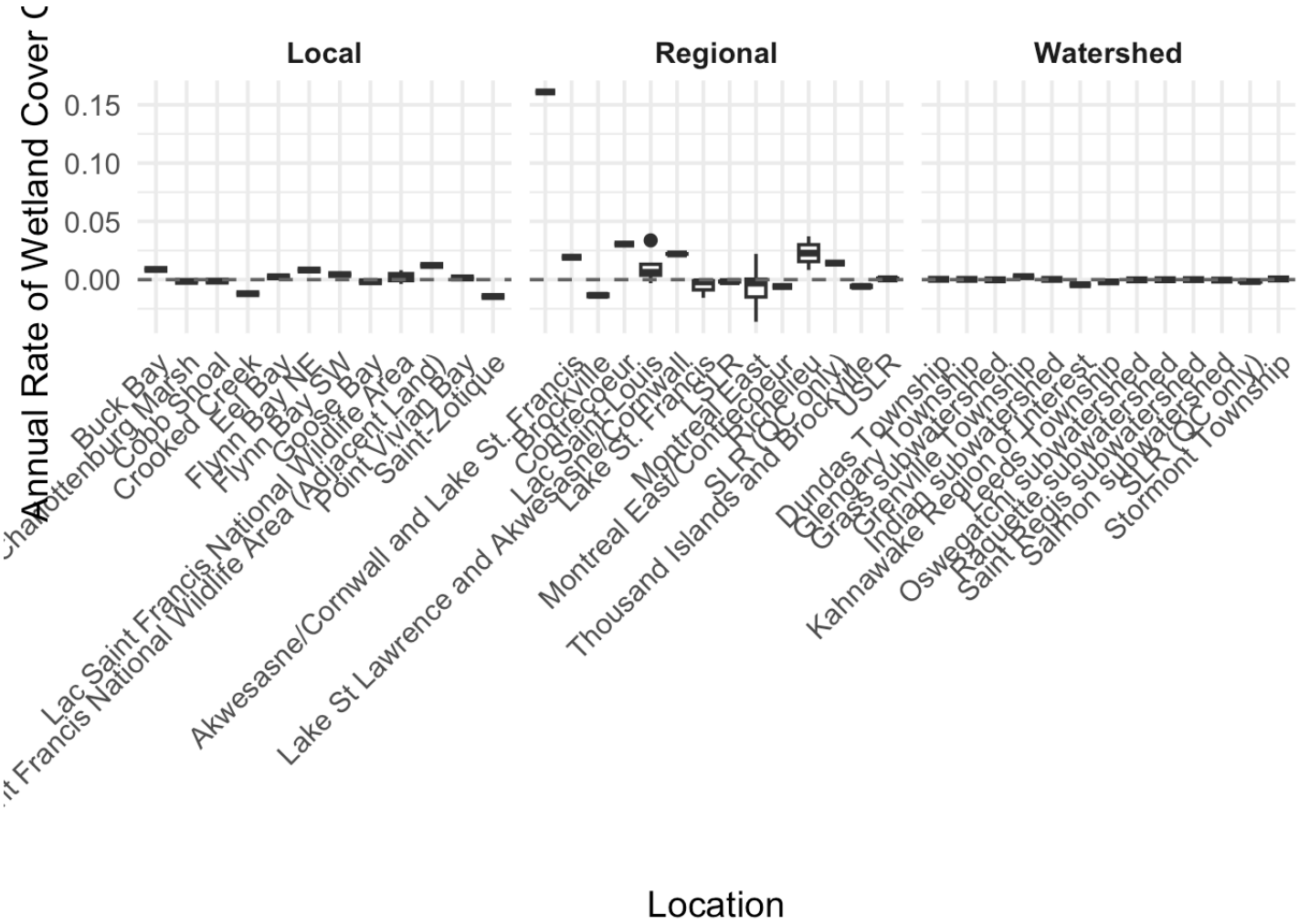
```
# Load libraries
library(readr)
library(dplyr)
library(ggplot2)

# ----------------------------
# 1. Load the dataset
# ----------------------------
wetland_df <- Wetland.Cover.Change.LRR

# ----------------------------
# 2. Keep only the 3 scale groups
# ----------------------------
scale_groups <- c("Local- Coastal Wetland",
                  "Regional - Coastal Wetlands",
                  "Watershed")

filtered_df <- wetland_df %>%
```

```
  filter(Scale %in% scale_groups)

# ----------------------------
# 3. Rename scale labels
# ----------------------------
filtered_df <- filtered_df %>%
  mutate(
    Scale_clean = case_when(
      Scale == "Local- Coastal Wetland" ~ "Local",
      Scale == "Regional - Coastal Wetlands" ~ "Regional",
      Scale == "Watershed" ~ "Watershed"
    )
  )

# ----------------------------
# 4. Remove missing values
# ----------------------------
filtered_df <- filtered_df %>%
  filter(!is.na(Annual.rate),
         !is.na(Sections))

# ----------------------------
# 5. Loop through each scale
# ----------------------------
for (s in unique(filtered_df$Scale_clean)) {

  cat("\n====================================\n")
  cat("Scale:", s, "\n")
  cat("====================================\n")

  scale_data <- filtered_df %>%
    filter(Scale_clean == s)

  # ----------------------------
  # ANOVA test
  # ----------------------------
  cat("\n--- ANOVA Results ---\n")
  anova_model <- aov(Annual.rate ~ Sections, data = scale_data)
  print(summary(anova_model))

  # ----------------------------
  # Kruskal-Wallis test (backup)
  # ----------------------------
  cat("\n--- Kruskal-Wallis Results ---\n")
  print(kruskal.test(Annual.rate ~ Sections, data = scale_data))

}
```

```
====================================
Scale: Regional
====================================

--- ANOVA Results ---
            Df   Sum Sq   Mean Sq F value   Pr(>F)
Sections    13 0.028776 0.0022136   11.13 7.14e-06 ***
Residuals   17 0.003381 0.0001989
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

--- Kruskal-Wallis Results ---

    Kruskal-Wallis rank sum test

data:  Annual.rate by Sections
Kruskal-Wallis chi-squared = 20.058, df = 13, p-value
= 0.09376


====================================
Scale: Local
====================================

--- ANOVA Results ---
            Df    Sum Sq   Mean Sq F value Pr(>F)
Sections    11 0.0006833 6.212e-05   0.877  0.691
Residuals    1 0.0000708 7.080e-05

--- Kruskal-Wallis Results ---

    Kruskal-Wallis rank sum test

data:  Annual.rate by Sections
Kruskal-Wallis chi-squared = 10.385, df = 11, p-value
= 0.4962


====================================
Scale: Watershed
====================================

--- ANOVA Results ---
            Df    Sum Sq   Mean Sq F value Pr(>F)
Sections    12 3.241e-05 2.701e-06    1.31  0.601
```

```
Residuals    1 2.060e-06 2.062e-06

--- Kruskal-Wallis Results ---

    Kruskal-Wallis rank sum test

data:  Annual.rate by Sections
Kruskal-Wallis chi-squared = 12.886, df = 12, p-value
= 0.3774
```

```
m=ggplot(filtered_df,
       aes(x = Sections, y = Annual.rate)) +

  geom_boxplot() +

  facet_wrap(~Scale_clean, scales = "free_x") +

  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "gray40") +

  labs(
    x = "Location",
    y = "Annual Rate of Wetland Cover Change"
  ) +

  theme_minimal(base_size = 14) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.text = element_text(face = "bold"),
    legend.position = "none"
  )
m
```


Local
Regional
Watershed
Annual Rate of Wetland Cover C
0.15
0.10
0.05
0.00
Buck Bay
Cottenburg Marsh
Cobb Shoal
Crooked Creek
Eel Bay
Flynn Bay NE
Flynn Bay SW
Goose Bay
Lac Saint Francis National Wildlife Area (Adjacent Land)
Point Vivian Bay
Saint-Zotique
Akwesasne/Cornwall and Lake St. Francis
Brockville
Contrecoeur
Lac Saint-Louis
Lake St Lawrence and Akwesasne/Cornwall
Lake St. Francis
LSLR
Montreal East
Montreal East/Contrecoeur
Richelieu
SLR (QC only)
Thousand Islands and Brockville
USLR
Dundas Township
Glengary Township
Grass subwatershed
Grenville Township
Indian subwatershed
Kahnawake Region of Interest
Leeds Township
Oswegatchi subwatershed
Raquette subwatershed
Saint Regis subwatershed
Salmon subwatershed
SLR (QC only)
Stormont Township
Location